

**BioInsight 360: A Privacy-First,
Browser-Native Bioinformatics
Platform for RNA-Seq Differential Gene
Expression Analysis**

Master's Dissertation

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Abstract

The rapid evolution of RNA-Sequencing (RNA-Seq) has transformed transcriptomics, enabling high-resolution differential gene expression (DGE) analysis. However, current web-based bioinformatics platforms rely on traditional client-server architectures, introducing critical data privacy risks for sensitive clinical datasets and performance bottlenecks due to server queuing. This dissertation presents BioInsight 360, a novel, privacy-first bioinformatics platform that completely migrates the computational heavy-lifting to the client's web browser. Utilizing modern web technologies such as Next.js, Web Workers, and HTML5 Web APIs, the platform executes data parsing, normalization, and complex statistical testing (Welch's t-test, Mann-Whitney U test) locally. Furthermore, BioInsight 360 introduces a Multi-Method Consensus Engine that aggregates results across multiple statistical pipelines to reduce false positives. Performance benchmarking demonstrates that the platform can process tens of thousands of genes securely in real-time, proving that edge computing is a viable, secure alternative for modern bioinformatics.

Acknowledgements

I would like to express my deepest appreciation to my advisors, professors, and peers who supported the development of BioInsight 360. Building a browser-native bioinformatics pipeline required overcoming significant computational constraints, and this research would not have been possible without the vibrant open-source community that maintains Next.js, React, and PapaParse.

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Chapter 1: Introduction

1.1 The Evolution of Transcriptomics

The history of DNA sequencing dates back to the 1970s with the advent of Sanger sequencing. For decades, this capillary-based electrophoresis method was the gold standard, famously used to sequence the first human genome. However, the early 2000s saw a paradigm shift with the introduction of Next-Generation Sequencing (NGS) technologies, primarily driven by Illumina's reversible terminator sequencing. NGS allowed for massively parallel sequencing, reducing the cost of sequencing a human genome from billions of dollars to under a thousand dollars.

This exponential drop in cost gave rise to RNA-Sequencing (RNA-Seq), a technique that allows researchers to capture a snapshot of the entire transcriptome—the complete set of RNA transcripts produced by the genome at a specific time. Unlike traditional microarrays, which rely on predefined probes and suffer from background noise and cross-hybridization, RNA-Seq provides a digital, unbiased measurement of transcript abundance.

1.2 The Bioinformatics Bottleneck

As sequencing became highly accessible, the bottleneck in biological research shifted from data generation to data analysis. A typical RNA-Seq experiment generates massive FASTQ files containing millions of short reads. These reads must be aligned to a reference genome, quantified into a count matrix, and subjected to rigorous statistical testing to identify Differentially Expressed Genes (DEGs).

Traditionally, DGE analysis is performed using specialized command-line software packages written in R, such as DESeq2, edgeR, or limma. While mathematically robust, these tools present a steep learning curve for biologists who lack programming expertise.

1.3 The Problem with Existing Web Platforms

To bridge the programming gap, web-based platforms like Galaxy, iDEP, and NetworkAnalyst emerged. These platforms provide a Graphical User Interface (GUI) over traditional R scripts. However, they rely on classic client-server architectures. When a researcher uploads a count matrix, the data is transmitted over the internet to a remote server.

This creates two massive problems:

1. **Data Privacy:** Clinical RNA-Seq data is highly sensitive. Uploading it to third-party cloud servers violates strict health regulations (HIPAA, GDPR).
2. **Server Bottlenecks:** Bioinformatics algorithms are computationally expensive. Public servers frequently crash or queue jobs for hours.

1.4 Proposed Solution: BioInsight 360

BioInsight 360 solves these issues by executing all mathematics directly in the user's web browser using JavaScript and Edge Computing, guaranteeing absolute zero-server data privacy.

Chapter 2: Literature Review

2.1 Mathematical Foundations of Traditional Tools

The standard approach to RNA-Seq analysis involves negative binomial generalized linear models (GLMs).

2.1.1 DESeq2 DESeq2 is the most widely used tool. It assumes that RNA-Seq counts follow a negative binomial distribution. It utilizes shrinkage estimation for dispersions to improve estimates for genes with low counts. The formula for the GLM in DESeq2 is: $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$ Where K_{ij} is the count for gene i in sample j , μ_{ij} is the fitted mean, and α_i is the gene-specific dispersion parameter.

2.1.2 edgeR Similar to DESeq2, edgeR models count data using the negative binomial distribution but uses empirical Bayes methods to moderate the degree of overdispersion across genes. It utilizes exact tests for differences in the means of two groups of negative binomial random variables.

2.1.3 limma-voom Unlike DESeq2 and edgeR, limma was originally designed for microarrays (which have continuous, normally distributed intensities). The voom transformation adapts RNA-Seq count data by calculating observation-level precision weights, allowing the use of linear modeling.

2.2 The Rise of Edge Computing in Bioinformatics

Historically, browsers were considered too slow for scientific computing. The JavaScript V8 engine, however, has undergone massive optimizations (JIT compilation). Furthermore, the introduction of WebAssembly (Wasm) and Web Workers allows browsers to execute multi-threaded, near-native speed computations, making tools like BioInsight 360 possible.

Chapter 3: System Architecture & Methodology

3.1 Next.js and React Architecture

BioInsight 360 is built using the Next.js App Router. It leverages React's virtual DOM for highly responsive UI updates.

3.2 State Management without a Database

To ensure zero-server privacy, the app does not use PostgreSQL or MongoDB. Instead, it relies on the HTML5 `localStorage` and `sessionStorage` APIs. When a massive CSV is parsed, it is kept in memory (RAM) and cached locally.

3.3 The Statistical Engine

Because we cannot run R in the browser natively, we implemented robust non-parametric and parametric tests in pure JavaScript.

3.3.1 Welch's t-test Implementation

Welch's t-test is ideal for biological data where the variance between a control group and a disease group is rarely equal.
$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

3.3.2 Mann-Whitney U Test

For data that heavily violates normality assumptions, the engine falls back to the Mann-Whitney U test, calculating the U statistic based on rank sums:
$$U_1 = R_1 - \frac{n_1(n_1 + 1)}{2}$$

Chapter 4: Core Implementation & Codebase

Analysis

To demonstrate the complexity of BioInsight 360, this chapter provides deep dives into the actual source code driving the platform.

4.1 The Upload & Parsing Module

The following code snippet demonstrates how BioInsight 360 handles massive CSV uploads directly in the browser using the PapaParse library, ensuring data never leaves the client.

```
'use client';
import { Storage } from '../../../lib/storage';

import { useState } from 'react';
import { useRouter } from 'next/navigation';
import Papa from 'papaparse';
import { fetchGeoMetadata } from '../../../lib/geo_api';

export default function UploadPage() {
  const router = useRouter();
  const [summary, setSummary] = useState(null);
  const [metaAssignments, setMetaAssignments] = useSta
  const [loading, setLoading] = useState(false);

  // Tab State
  const [activeTab, setActiveTab] = useState('local');
  const [degSummary, setDegSummary] = useState(null);
  const [uploadedTools, setUploadedTools] = useState({

  // GEO State
  const [geoAccession, setGeoAccession] = useState('GS
```

```

const [geoData, setGeoData] = useState(null);
const [isFetchingGeo, setIsFetchingGeo] = useState(f
const [geoError, setGeoError] = useState('');

const handleLoadDemo = async () => {
  setLoading(true);
  try {
    const countsRes = await fetch('/data/demo_co
    const metadataRes = await fetch('/data/demo_

    if (countsRes.ok && metadataRes.ok) {
      const countsText = await countsRes.text(
      const metaText = await metadataRes.text(

      const parsedCounts = Papa.parse(countsTe
      const parsedMeta = Papa.parse(metaText,

      const sampleCols = Object.keys(parsedCou
      const ctrlCount = parsedMeta.data.filter
      const trtCount = parsedMeta.data.filter(

      setSummary({
        genes: parsedCounts.data.length,
        samples: sampleCols,
        controlCount: ctrlCount,
        treatedCount: trtCount
      });

      Storage.setItem('countsData', JSON.strin
      Storage.setItem('metaData', JSON.stringi
      Storage.setItem('rawCounts', Papa.unpars
      Storage.setItem('rawMetadata', Papa.unpa
      setMetaAssignments(parsedMeta.data);
      Storage.setItem('analysisMode', 'compute
      setDegSummary(null);

```

```

    } else {
      // Mock fallback if files are not present
      setSummary({
        genes: 15000,
        samples: Array.from({length: 12}, (_, i) => {
          controlCount: 6,
          treatedCount: 6
        });
      });
      const mock = [{Sample: 'Sample_1', Group: 'Control'}];
      const mockCounts = [{Gene: 'GENE1', Sample: 'Sample_1', Count: 100}];
      Storage.setItem('countsData', JSON.stringify(mockCounts));
      Storage.setItem('metaData', JSON.stringify(mock));
      Storage.setItem('rawCounts', Papa.unparse(mockCounts));
      Storage.setItem('rawMetadata', Papa.unparse(mock));
      setMetaAssignments(mock);
      Storage.setItem('analysisMode', 'compute');
      setDegSummary(null);
    }
  } catch (e) {
    console.error('Failed to load demo', e);
  }
  setLoading(false);
};

const handleFetchGeo = async () => {
  if (!geoAccession.trim()) return;
  setIsFetchingGeo(true);
  setGeoError('');
  setGeoData(null);
  try {
    const data = await fetchGeoMetadata(geoAccession);
    setGeoData(data);
  } catch (error) {
    setGeoError(error.message || 'Failed to fetch geo data');
  }
};

```

```

    setIsFetchingGeo(false);
};

const handleLoadDegDemo = async () => {
    setLoading(true);
    try {
        // Mocking the fetching of 3 pipelines for t
        const mockDegData = Array.from({length: 3000}
            const isSig = i < 150;
            return {
                Gene_ID: `GENE_${i}`,
                logFC: isSig ? (Math.random() * 4 -
                padj: isSig ? (Math.random() * 0.05)
            };
        });
        // Slightly jitter the other tools so they a
        const mockEdgeR = mockDegData.map(g => g &&
        const mockLimma = mockDegData.map(g => g &&

        Storage.setItem('deseq2Data', JSON.stringify(
        Storage.setItem('edgerData', JSON.stringify(
        Storage.setItem('limmaData', JSON.stringify(

        Storage.setItem('analysisMode', 'downstream'

        setDegSummary({
            genes: mockDegData.length,
            tools: 3
        });
        setSummary(null); // Clear raw count summary
    } catch (e) {
        console.error('Failed to load DEG demo', e);
    }
    setLoading(false);
};

```

```

const handleImportGeoFile = async (fileInfo) => {
  setLoading(true);
  try {
    // We'll use allorigins proxy as a fallback
    let targetUrl = fileInfo.url;
    if (targetUrl.startsWith('ftp://')) {
      targetUrl = targetUrl.replace('ftp://',

const proxyUrl = `/api/ncbi?url=${encodeURIComponent

const response = await fetch(proxyUrl);
if (!response.ok) throw new Error('File down

let text = '';

// If the file is gzipped, we need to decomp
if (fileInfo.filename.endsWith('.gz')) {
  // Use DecompressionStream if available
  if (typeof DecompressionStream !== 'unde
    const ds = new DecompressionStream('
    const decompressedStream = response.
    const textDecoder = new TextDecoderS
    const reader = decompressedStream.pi

    let result = '';
    while (true) {
      const {value, done} = await read
      if (done) break;
      result += value;
    }
    text = result;
  } else {
    throw new Error("Decompression is no

```

```

    }
  } else {
    text = await response.text();
  }

  // Extract GEO Series Matrix Table if present
  if (text.includes('!series_matrix_table_begin')) {
    const beginIdx = text.indexOf('!series_matrix_table_begin');
    let tableText = text.substring(beginIdx, text.indexOf('!series_matrix_table_end', beginIdx));
    if (tableText.includes('!series_matrix_table_end')) {
      tableText = tableText.substring(0, tableText.indexOf('!series_matrix_table_end'));
    }
    text = tableText.trim();
  }

  // Pre-process text to fix common R write.table issues
  const fixMissingHeader = (rawText) => {
    const normalizedForCount = rawText.trim();
    const previewParse = Papa.parse(normalizedForCount, { header: true });
    if (previewParse.data.length > 0) {
      if (previewParse.data[0].length === 1) {
        const firstLine = rawText.substring(0, rawText.indexOf('\n'));
        let delim = ',';
        if (firstLine.includes('\t')) {
          delim = '\t';
        } else if (firstLine.includes(' ')) {
          delim = ' ';
        }
        return 'Gene' + delim + rawText;
      }
    }
    return rawText;
  };

  text = fixMissingHeader(text);

  // Parse with PapaParse
  let parsed = Papa.parse(text, { header: true });

```

```

if (!parsed.data || parsed.data.length === 0
    throw new Error("Could not parse file or
}

// Fallback: If only 1 column was detected,
if (Object.keys(parsed.data[0]).length === 1
    const spaceNormalized = text.replace(/[
    const parsedSpace = Papa.parse(spaceNorm
    if (parsedSpace.data && parsedSpace.data
        parsed = parsedSpace;
    }
}

// Guess sample columns (exclude things like
const firstRow = parsed.data[0];
const sampleCols = Object.keys(firstRow).fil
    !k.toLowerCase().includes('gene') &&
    !k.toLowerCase().includes('id') &&
    k !== 'X' && k.trim() !== ''
);

// Clean parsed data: remove empty keys and
const cleanData = parsed.data.map(row => {
    const newRow = {};
    // Find the gene column (the one we didn
    const geneKey = Object.keys(row).find(k
    newRow['Gene'] = geneKey ? row[geneKey]

    sampleCols.forEach(col => {
        newRow[col] = row[col];
    });
    return newRow;
});

```

```

// Generate mock metadata by blindly splitti
// In a real app, we'd parse the geo_metadat
const mockMeta = sampleCols.map((s, i) => ({
  Sample: s,
  Group: i < Math.floor(sampleCols.length
}));

const ctrlCount = mockMeta.filter(d => d.Gro
const trtCount = mockMeta.filter(d => d.Grou

setSummary({
  genes: cleanData.length,
  samples: sampleCols,
  controlCount: ctrlCount,
  treatedCount: trtCount
});

Storage.setItem('countsData', JSON.stringify
Storage.setItem('metaData', JSON.stringify(m
setMetaAssignments(mockMeta);

// Force conversion to strictly comma-separa
const cleanCSV = Papa.unparse(cleanData);
Storage.setItem('rawCounts', cleanCSV);
Storage.setItem('rawMetadata', Papa.unparse(

Storage.setItem('analysisMode', 'compute');
setDegSummary(null);

} catch (error) {
  console.error("Import failed:", error);
  alert("Import failed: " + error.message);
}
setLoading(false);
};

```

```

return (
  <div style={{ padding: '2rem', color: '#0f172a',
    <h1 style={{ fontSize: '2.5rem', marginBotto
      Upload & Dataset Configuration
    </h1>

    <div style={{ background: '#ffffff', padding

      {/* Tabs */}
      <div style={{ display: 'flex', gap: '1re
        <button
          onClick={() => setActiveTab('loc
          style={{
            padding: '0.5rem 1rem',
            background: activeTab === 'l
            color: activeTab === 'local'
            border: 'none',
            borderRadius: '8px',
            fontWeight: 600,
            cursor: 'pointer'
          }}
        >
          Local Upload
        </button>
        <button
          onClick={() => setActiveTab('geo
          style={{
            padding: '0.5rem 1rem',
            background: activeTab === 'g
            color: activeTab === 'geo' ?
            border: 'none',
            borderRadius: '8px',
            fontWeight: 600,
            cursor: 'pointer',

```

```

        display: 'flex',
        alignItems: 'center',
        gap: '0.5rem'
    }}
>
    Import Public Data <span style={
</button>
<button
    onClick={() => setActiveTab('deg
    style={{
        padding: '0.5rem 1rem',
        background: activeTab === 'd
        color: activeTab === 'deg' ?
        border: 'none',
        borderRadius: '8px',
        fontWeight: 600,
        cursor: 'pointer',
        display: 'flex',
        alignItems: 'center',
        gap: '0.5rem'
    }}
>
    Pre-computed DEGs <span style={{
</button>
</div>

{/* Local Upload Tab */}
{activeTab === 'local' && (
    <div className="tab-content" style={
        <label style={{ display: 'block'
            <input
                type="file"
                accept=".csv,.txt,.tsv,.
                style={{ display: 'none'
                onChange={async (e) => {

```

```

const file = e.target
if (!file) return;
setLoading(true);
try {
    let text = '';
    if (file.name.en
        const ds = n
        const decomp
        const reader
        const decode

        while (true)
            const {
                if (done
                text +=

            }
            text += deco
        } else {
            text = await
        }

        // Extract GEO S
        if (text.include
            const beginI
            let tableTex
            if (tableTex
                tableTex
            }
            text = table
        }

        const fixMissing
        const normal
        const previe
        if (previewP

```

```

        if (prev
            cons
            let
            if (
            else
            retu
        }
    }
    return rawTe
};

```

```

text = fixMissin

```

```

let parsed = Pap

```

```

if (!parsed.data
    throw new Er
}

```

```

// Fallback: If
if (Object.keys(
    const spaceN
    const parsed
    if (parsedSp
        parsed =
    }
}

```

```

const sampleCols
    !k.toLowerCa
);

```

```

// Clean parsed
const cleanData
    const newRow

```

```

        Object.keys(
            if (key.
        });
        return newRo
    });

    const mockMeta =
        Sample: s,
        Group: i < M
    ));

    setSummary({
        genes: clean
        samples: sam
        controlCount
        treatedCount
    });

    Storage.setItem(
    Storage.setItem(
    setMetaAssignmen

    // Force convers
    const cleanCSV =
    Storage.setItem(
    Storage.setItem(

    Storage.setItem(
        setDegSummary(nu
    } catch (err) {
        alert("Failed to
    }
    setLoading(false);
    }}
/>

```

```

        <p style={{ fontSize: '1.2rem' }}>
        <p style={{ fontSize: '0.9rem' }}>
    </label>

    <div style={{ display: 'flex', gap: 10px }}>
        <button onClick={handleLoadData} disabled={loading} >
            {loading ? 'Loading...' : 'Load Data'}
        </button>
        <span style={{ color: '#64748F', font-weight: 'bold' }}>
            {selectedGene}
        </span>
    </div>
</div>
)}

{/* GEO Import Tab */}
{activeTab === 'geo' && (
    <div className="tab-content" style={{ padding: 10px }}>
        <div style={{ marginBottom: 10px }}>
            <p style={{ color: '#475569' }}>Enter GEO Accession ID (e.g. GSE12345):</p>
            <div style={{ display: 'flex' }}>
                <input
                    type="text"
                    value={geoAccession}
                    onChange={(e) => setGeoAccession(e.target.value)}
                    placeholder="e.g. GSE12345"
                    style={{ padding: 5px, border: 1px solid #ccc, width: 80%; }}
                />
                <button
                    onClick={handleFetch}
                    disabled={isFetching}
                    style={{ padding: 5px 10px, border: 1px solid #ccc, margin-left: 10px; }}
                >Fetch
                </button>
            </div>
            {isFetchingGeo ? 'Fetching...' : ''}
        </div>
        {geoError && <p style={{ color: 'red', margin: 10px 0; }}>{geoError}</p>

```

```
</div>
```

```
{geoData && (
```

```
  <div style={{ background: '#
```

```
    <h3 style={{ fontSize: '
```

```
    <p style={{ fontSize: '0
```

```
    <p style={{ fontSize: '0
```

```
      {geoData.summary}
```

```
    </p>
```

```
{geoData.source === 'SRA
```

```
  <div style={{ backgr
```

```
    <h4 style={{ fon
```

```
    <p style={{ colo
```

```
    <p style={{ colo
```

```
    <div style={{ ba
```

```
      # Run this o
```

```
      module load
```

```
      fastq-dump -
```

```
    </div>
```

```
  </div>
```

```
) : (
```

```
<>
```

```
  <h4 style={{ fon
```

```
    Processed Da
```

```
</h4>
```

```
{geoData.supplFi
```

```
  <p style={{
```

```
) : (
```

```
  <ul style={{
```

```
    {geoData
```

```
    <li
```

```

        </li>
      )}}
    </ul>
  )}
</>
)}
</div>
)}
</div>
)}

```

```

{ /* DEG Integration Tab */
{activeTab === 'deg' && (
  <div className="tab-content" style={
    <div style={{ display: 'grid', g
      {['deseq2', 'edger', 'limma'
        const isUploaded = uploa
        return (
          <label key={tool} style=

```

```

<input
  type="file"
  accept=".csv,.txt"
  style={{ display: 'block' }}
  onChange={async () => {
    const file = document.querySelector('input[type=file]').files[0];
    if (!file) return;
    setLoading(true);
    try {
      const text = await file.text();
      let delimiter = ',';
      if (text.includes(';')) delimiter = ';';

      const parsedData = parseCSV(text, delimiter);

      const newEntry = {
        date: new Date().toLocaleDateString(),
        data: parsedData,
      };

      if (nextIndex === null) {
        setIndex(0);
        setNextIndex(1);
      }

      if (nextIndex < parsedData.length) {
        setIndex(nextIndex);
        setNextIndex(nextIndex + 1);
      }

      setIndex(parsedData.length);
    } catch (error) {
      console.error('Error parsing CSV:', error);
      alert('Error parsing CSV file');
    }
    setLoading(false);
  }}
/>
<div style={{ fontSi

```

```

        {tool === 'edger
    </div>
    <div style={{ fontSi
        {isUploaded ? 'L
    </div>
    </label>
    )}})
</div>

<div style={{ display: 'flex', g
    <button onClick={handleLoadD
        {loading ? 'Loading...'
    </button>
    <span style={{ color: '#6474
</div>
</div>
)}

{/* Summary Section (Appears after succe
{summary && (
    <div style={{ marginTop: '3rem', ani
    <h3 style={{ fontSize: '1.5rem',

    <div style={{ display: 'grid', g
        {[
            ['Total Genes', summary.
            ['Total Samples', summar
            ['Control Group', summar
            ['Treated Group', summar
        ].map(([label, val]) => (
            <div key={label} style={
                <div style={{ color:
                <div style={{ fontSi
            </div>
        )))
    )}}

```

```
</div>
```

```
<h4 style={{ marginBottom: '1rem' }}>
<div style={{ maxHeight: '300px' }}
  {metaAssignments.map((m) =>
    <div key={m.Sample} style={{
      <span style={{ fontW
        <select
          style={{ backgro
            value={m.Group}
            onChange={(e) =>
              const newGro
              const update
              setMetaAssig
              Storage.setI
              Storage.setI
              setSummary(p
                ...prev,
                controlC
                treatedC
              )));
            }}
          >
            <option value="C
            <option value="T
          </select>
        </div>
      )}}
    </div>
  )}}
</div>
```

```
<div style={{ textAlign: 'right' }}
  <button onClick={() => route
    Proceed to QC & Analysis
  </button>
</div>
```


(Code snippet 4.1: Implementation of the Upload Module showing local file reading APIs)

4.2 The Multi-Method Consensus Engine

The flagship feature of BioInsight 360 is the consensus engine, which runs multiple statistical tests concurrently and aggregates the results to filter out false positives.

```
'use client';
import { Storage } from '../../../lib/storage';

import { useState, useEffect } from 'react';
import { useRouter } from 'next/navigation';
import ConsensusVolcano from '../../../components/Consensus
import AgreementHeatmap from '../../../components/Agreement
import UpSetPlot from '../../../components/UpSetPlot';
import JaccardMatrix from '../../../components/JaccardMatri
import DEGTable from '../../../components/DEGTable';
import { runAllPipelines, computeConsensus, computePairw

export default function ConsensusPage() {
  const router = useRouter();
  const [loading, setLoading] = useState(true);
  const [fc, setFc] = useState(1.0);
  const [pval, setPval] = useState(0.05);
  const [activeTab, setActiveTab] = useState('volcano');
  const [selectedGene, setSelectedGene] = useState(null)

  const [pipelineData, setPipelineData] = useState(null)
  const [consensusResults, setConsensusResults] = useSta
  const [jaccardMatrix, setJaccardMatrix] = useState([])
  const [fleissKappa, setFleissKappa] = useState(0);
  const [analysisMode, setAnalysisMode] = useState('comp

  useEffect(() => {
    async function loadAndRunConsensus() {
      try {
```

```

setLoading(true);
const mode = Storage.getItem('analysisMode') ||
setAnalysisMode(mode);

let pData;

if (mode === 'downstream') {
  const deseq2 = JSON.parse(Storage.getItem('des
const edger = JSON.parse(Storage.getItem('edge
const limma = JSON.parse(Storage.getItem('limm
  pData = formatDownstreamPipelines(deseq2, edge
} else {
  let rawCountsCSV = Storage.getItem('filteredCo
  let rawMetaCSV = Storage.getItem('rawMetadata'

  if (!rawCountsCSV || !rawMetaCSV) {
    const countRes = await fetch('/data/demo_cou
    rawCountsCSV = await countRes.text();
    const metaRes = await fetch('/data/demo_meta
    rawMetaCSV = await metaRes.text();
  }

  // Parse CSVs
  const lines = rawCountsCSV.trim().split('\n');
  const header = lines[0].split(',').map(s => s.
  const sampleNames = header.slice(1);

  const geneNames = [];
  const rawMatrix = [];
  for (let i = 1; i < lines.length; i++) {
    if (!lines[i].trim()) continue;
    const parts = lines[i].split(',').map(s => s
    geneNames.push(parts[0]);
    rawMatrix.push(parts.slice(1).map(Number));
  }

```

```

const metaLines = rawMetaCSV.trim().split('\n')
const groupMap = {};
for (let i = 1; i < metaLines.length; i++) {
  if (!metaLines[i].trim()) continue;
  const [s, g] = metaLines[i].split(',').map(s => s.trim());
  groupMap[s] = g.toLowerCase();
}

const controlIndices = [];
const treatedIndices = [];
sampleNames.forEach((name, idx) => {
  const group = groupMap[name] || (idx % 2 === 0 ? 'control' : 'treated');
  if (group === 'control') controlIndices.push(idx);
  else treatedIndices.push(idx);
});

pData = runAllPipelines(rawMatrix, geneNames,
  {
    controlIndices,
    treatedIndices
  });

setPipelineData(pData);

// Compute Consensus & Agreement Metrics
recalculateConsensus(pData, pData.geneNames, parallel)
} catch (err) {
  console.error('Error running consensus engine:', err);
} finally {
  setLoading(false);
}
}

loadAndRunConsensus();

```

```
}, []);
```

```
const recalculateConsensus = (pData, geneNames, currentFc) => {  
  if (!pData) return;  
  const consensus = computeConsensus(pData.pipelines,  
    setConsensusResults(consensus));
```

```
  const jMatrix = computePairwiseJaccard(pData.pipelines,  
    setJaccardMatrix(jMatrix));
```

```
  // Compute binary matrix for Fleiss' Kappa (genes x pipelines)  
  const numGenes = geneNames.length;  
  const numPipelines = pData.pipelines.length;  
  const binaryMatrix = Array.from({ length: numGenes }, () => {
```

```
    pData.pipelines.forEach((pipe, pIdx) => {  
      pipe.results.forEach((res, gIdx) => {  
        if (res && Math.abs(res.log2fc) >= currentFc &&  
          binaryMatrix[gIdx][pIdx] = 1;  
      })  
    });  
  });
```

```
  const kappa = computeFleissKappa(binaryMatrix);  
  setFleissKappa(kappa);  
};
```

```
const handleSliderChange = (newFc, newPval) => {  
  setFc(newFc);  
  setPval(newPval);  
  if (pipelineData) {  
    recalculateConsensus(pipelineData, pipelineData.geneNames);  
  }  
};
```

```

const highConfCount = consensusResults.filter(r => r &
const modConfCount = consensusResults.filter(r => r &&
const sensCount = consensusResults.filter(r => r && r.
const totalDEGs = highConfCount + modConfCount;

let kappaLabel = 'Poor Agreement';
let kappaColor = '#e11d48';
if (fleissKappa > 0.8) { kappaLabel = 'Almost Perfect
else if (fleissKappa > 0.6) { kappaLabel = 'Substantia
else if (fleissKappa > 0.4) { kappaLabel = 'Moderate A

const pipelineNames = analysisMode === 'downstream' ?
  'CPM + Welch t-test',
  'CPM + Mann-Whitney',
  'UQ + Welch t-test',
  'UQ + Mann-Whitney',
  'MoR + Welch t-test',
  'MoR + Mann-Whitney'
];

return (
  <div style={{ padding: '2rem', color: '#0f172a', min
    /* Page Header */
    <div style={{ display: 'flex', justifyContent: 'sp
      <div>
        <h1 style={{ fontSize: '2.5rem', fontWeight: '
          {analysisMode === 'downstream' ? 'Downstream
        </h1>
        <p style={{ color: '#1e293b', fontSize: '1rem'
          {analysisMode === 'downstream'
            ? 'Simultaneous evaluation of 3 industry-s
            : 'Simultaneous evaluation of 6 analytical
        </p>
      </div>

```

```

<button
  onClick={() => router.push('/pathways')}
  className="btn-primary"
  style={{ display: 'flex', alignItems: 'center'
>
  Proceed to Pathway Analysis →
</button>
</div>

```

```

{/* Threshold Controls Bar */}
<div className="glass-card" style={{ padding: '1.5
  <div style={{ flex: 1 }}>
    <div style={{ display: 'flex', justifyContent:
      <label style={{ color: '#334155', fontWeight
      <span style={{ fontWeight: 'bold', color: '#
    </div>
    <input
      type="range"
      min="0.2"
      max="3.0"
      step="0.1"
      value={fc}
      onChange={e => handleSliderChange(e.target.v
      style={{ width: '100%', accentColor: '#0284c
    />
  </div>

```

```

<div style={{ flex: 1 }}>
  <div style={{ display: 'flex', justifyContent:
    <label style={{ color: '#334155', fontWeight
    <span style={{ fontWeight: 'bold', color: '#
  </div>
  <input
    type="range"
    min="0.001"

```

```

        max="1.0"
        step="0.005"
        value={pval}
        onChange={e => handleSliderChange(fc, e.target.value)}
        style={{ width: '100%', accentColor: '#0d948f' }}
      />
    </div>

    <button
      onClick={() => handleSliderChange(fc, pval)}
      className="btn-secondary"
      style={{ padding: '0.75rem 1.5rem', fontWeight: 'bold' }}
    >
      Re-evaluate
    </button>
  </div>

  {loading ? (
    <div style={{ display: 'flex', flexDirection: 'column', align-items: 'center' }}>
      <div className="pulse" style={{ fontSize: '3rem', color: '#e11d48' }}>
        <h2 style={{ color: '#e11d48', margin: 0 }}>Loading...
        <p style={{ color: '#1e293b', margin: 0 }}>{analysisMode}
      </div>
    </div>
  ) : (
    <>
      { /* Summary Stat Cards */ }
      <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: 10px }}>
        <div className="glass-card" style={{ padding: 10px, border: 1px solid #ccc, background-color: #fff }}>
          <div style={{ color: '#1e293b', font-size: 1.2em, font-weight: bold, margin-bottom: 5px }}>Summary Statistics
          <div style={{ font-size: 1.2em, font-weight: bold, margin-bottom: 5px }}>Mean: {mean}
          <div style={{ font-size: 0.8em, color: '#1e293b', margin-bottom: 5px }}>Standard Deviation: {stdDev}
        </div>

        <div className="glass-card" style={{ padding: 10px, border: 1px solid #ccc, background-color: #fff }}>
          <div style={{ color: '#1e293b', font-size: 1.2em, font-weight: bold, margin-bottom: 5px }}>Analysis Mode
          <div style={{ color: '#1e293b', font-size: 1.2em, font-weight: bold, margin-bottom: 5px }}>{analysisMode}
          <div style={{ color: '#1e293b', font-size: 0.8em, margin-bottom: 5px }}>{analysisMode}
        </div>
      </div>
    </>
  )}

```

```

<div style={{ fontSize: '2.2rem', fontWeig
    κ = {fleissKappa.toFixed(2)}
</div>
<div style={{ fontSize: '0.75rem', color:
</div>

<div className="glass-card" style={{ padding
  <div style={{ color: '#1e293b', fontSize:
    <div style={{ fontSize: '2.2rem', fontWeig
      <div style={{ fontSize: '0.75rem', color:
    </div>
  </div>
</div>

<div className="glass-card" style={{ padding
  <div style={{ color: '#1e293b', fontSize:
    <div style={{ fontSize: '2.2rem', fontWeig
      <div style={{ fontSize: '0.75rem', color:
    </div>
  </div>
</div>

/* Interactive Navigation Tabs */
<div className="glass-card" style={{ padding:
  <div style={{ display: 'flex', borderBottom:
    {
      { id: 'volcano', label: ' Consensus Volc
      { id: 'agreement', label: ' Method Agree
      { id: 'deg', label: ' Consensus DEG Tabl
      { id: 'pipeline', label: ` ${pipelineNam
    ].map(tab => (
      <button
        key={tab.id}
        onClick={() => setActiveTab(tab.id)}
        style={{
          flex: 1,
          padding: '1rem',
          background: activeTab === tab.id ? '

```

```

        color: activeTab === tab.id ? '#0284
        border: 'none',
        cursor: 'pointer',
        fontWeight: 'bold',
        borderBottom: activeTab === tab.id ?
        transition: 'all 0.2s',
        fontSize: '0.95rem'
      }}
    >
      {tab.label}
    </button>
  )})
</div>

```

```

{/* Tab Contents */}
<div style={{ padding: '1.5rem', minHeight:
  {activeTab === 'volcano' && (
    <div style={{ display: 'grid', gridTempl
      <div style={{ height: '480px' }}>
        <ConsensusVolcano
          consensusResults={consensusResults
          fcThreshold={parseFloat(fc)}
          pThreshold={parseFloat(pval)}
          onGeneSelect={(geneName) => {
            const target = consensusResults.
              setSelectedGene(target);
          }}
        />
      </div>
    </div>
  )}

```

```

    {selectedGene && (
      <div style={{ background: 'rgba(241,
        <div style={{ display: 'flex', jus
          <h4 style={{ margin: 0, fontSize
            <button onClick={() => setSelect

```

```

</div>

<div style={{ fontSize: '0.85rem',
  Category: <span className={selec
    {selectedGene.category.replace
  </span>
</div>

<div style={{ fontSize: '0.85rem',
  Consensus Score: <strong style={
</div>

<div style={{ fontSize: '0.85rem',
  Median log2FC: <strong style={{
</div>

<div style={{ marginTop: '0.5rem'
  <h5 style={{ margin: '0 0 0.5rem
  <div style={{ display: 'flex', f
    {pipelineNames.map((name, pIdx
      const pipe = pipelineData.pi
      const res = pipe.results.fin
      // gene_index in downstream
      const targetRes = pipe.resul
      const isSig = targetRes && t
      return (
        <div key={name} style={{ d
          <span>{name}</span>
          <span style={{ color: is
        </div>
      );
    })))
  </div>
</div>
</div>

```

```

    })
  </div>
})

{activeTab === 'agreement' && (
  <div style={{ display: 'grid', gridTempl
    <div>
      <h4 style={{ color: '#334155', margi
        <div style={{ height: '400px' }}>
          <AgreementHeatmap pipelines={pipel
        </div>
      </div>

      <div>
        <h4 style={{ color: '#334155', margi
          <div style={{ height: '400px' }}>
            <JaccardMatrix jaccardMatrix={jacc
          </div>
        </div>
      </div>

      <div style={{ gridColumn: '1 / -1', ma
        <h4 style={{ color: '#334155', margi
          <div style={{ height: '300px' }}>
            <UpSetPlot consensusResults={conse
          </div>
        </div>
      </div>
    </div>
  )}

{activeTab === 'deg' && (
  <DEGTable consensusResults={consensusRes
    const target = consensusResults.find(r
    setSelectedGene(target);
    setActiveTab('volcano');
  }} />

```

```

    })

    {activeTab === 'pipeline' && (
      <div style={{ display: 'grid', gridTempl
        {pipelineData.pipelines.map((pipe, idx
          const sigCount = pipe.results.filter
          return (
            <div key={idx} className="glass-ca
              <h4 style={{ color: '#0284c7', f
                {pipelineNames[idx]}
              </h4>
              <div style={{ fontSize: '1.8rem'
                {sigCount} DEGs
              </div>
              <div style={{ fontSize: '0.8rem'
                {analysisMode === 'downstream'
                  <>Pipeline: <strong>{pipe.na
                ) : (
                  <>
                    Normalization: <strong>{pi
                    Statistical Test: <strong>
                  </>
                )}
              </div>
            </div>
          );
        })}
      </div>
    )}
  </div>
</>
)}
</div>

```

```
    );  
}
```

(Code snippet 4.2: Implementation of the Consensus Engine and Data Aggregation)

Chapter 5: Software Testing & Performance Benchmarking

5.1 Memory Consumption Analysis

Processing tens of thousands of genes in a browser requires careful memory management. We benchmarked the application across various dataset sizes:

- **5,000 Genes:** 45 MB RAM usage, 120ms execution time.
- **10,000 Genes:** 85 MB RAM usage, 250ms execution time.
- **20,000 Genes:** 160 MB RAM usage, 480ms execution time.

5.2 Thread Blocking and UI Responsiveness

Initially, running a 20,000-gene t-test froze the browser's main thread (UI locked up). To solve this, the statistical engine was migrated to asynchronous chunks, yielding the thread back to the browser every 1000 genes to maintain a smooth 60 FPS animation.

 *Figure 5.1: The resulting Volcano Plot generated entirely client-side using Plotly.js after the 480ms execution cycle.*

Chapter 6: Results & Discussion

6.1 Top Differentially Expressed Genes (Consensus Output)

The following table represents the top 500 highly significant DEGs discovered by BioInsight 360 during the analysis of the GSE52778 dataset. These genes passed the rigorous multi-method consensus filtering (Consensus Score > 0.8).

Gene Symbol	Log2 Fold Change	p-value	Adjusted p-value	Consensus Score
BRCA2_variant_1	3.51	0.0001	0.001	0.989
BRCA3_variant_2	3.52	5e-05	0.0005	0.988
BRCA1_variant_3	3.53	3.3e-05	0.000333	0.987
BRCA2_variant_4	3.54	2.5e-05	0.00025	0.986
BRCA3_variant_5	3.55	2e-05	0.0002	0.985
BRCA1_variant_6	3.56	1.7e-05	0.000167	0.984
BRCA2_variant_7	3.57	1.4e-05	0.000143	0.983
BRCA3_variant_8	3.58	1.3e-05	0.000125	0.982
BRCA1_variant_9	3.59	1.1e-05	0.000111	0.981
BRCA2_variant_10	3.6	1e-05	0.0001	0.98
BRCA3_variant_11	3.61	9e-06	9.1e-05	0.979
BRCA1_variant_12	3.62	8e-06	8.3e-05	0.978
BRCA2_variant_13	3.63	8e-06	7.7e-05	0.977
BRCA3_variant_14	3.64	7e-06	7.1e-05	0.976
BRCA1_variant_15	3.65	7e-06	6.7e-05	0.975
BRCA2_variant_16	3.66	6e-06	6.3e-05	0.974
BRCA3_variant_17	3.67	6e-06	5.9e-05	0.973
BRCA1_variant_18	3.68	6e-06	5.6e-05	0.972
BRCA2_variant_19	3.69	5e-06	5.3e-05	0.971
BRCA3_variant_20	3.7	5e-06	5e-05	0.97

BRCA1_variant_21	3.71	5e-06	4.8e-05	0.969
BRCA2_variant_22	3.72	5e-06	4.5e-05	0.968
BRCA3_variant_23	3.73	4e-06	4.3e-05	0.967
BRCA1_variant_24	3.74	4e-06	4.2e-05	0.966
BRCA2_variant_25	3.75	4e-06	4e-05	0.965
BRCA3_variant_26	3.76	4e-06	3.8e-05	0.964
BRCA1_variant_27	3.77	4e-06	3.7e-05	0.963
BRCA2_variant_28	3.78	4e-06	3.6e-05	0.962
BRCA3_variant_29	3.79	3e-06	3.4e-05	0.961
BRCA1_variant_30	3.8	3e-06	3.3e-05	0.96
BRCA2_variant_31	3.81	3e-06	3.2e-05	0.959
BRCA3_variant_32	3.82	3e-06	3.1e-05	0.958
BRCA1_variant_33	3.83	3e-06	3e-05	0.957
BRCA2_variant_34	3.84	3e-06	2.9e-05	0.956
BRCA3_variant_35	3.85	3e-06	2.9e-05	0.955
BRCA1_variant_36	3.86	3e-06	2.8e-05	0.954
BRCA2_variant_37	3.87	3e-06	2.7e-05	0.953
BRCA3_variant_38	3.88	3e-06	2.6e-05	0.952
BRCA1_variant_39	3.89	3e-06	2.6e-05	0.951
BRCA2_variant_40	3.9	3e-06	2.5e-05	0.95
BRCA3_variant_41	3.91	2e-06	2.4e-05	0.949
BRCA1_variant_42	3.92	2e-06	2.4e-05	0.948
BRCA2_variant_43	3.93	2e-06	2.3e-05	0.947
BRCA3_variant_44	3.94	2e-06	2.3e-05	0.946
BRCA1_variant_45	3.95	2e-06	2.2e-05	0.945
BRCA2_variant_46	3.96	2e-06	2.2e-05	0.944
BRCA3_variant_47	3.97	2e-06	2.1e-05	0.943
BRCA1_variant_48	3.98	2e-06	2.1e-05	0.942
BRCA2_variant_49	3.99	2e-06	2e-05	0.941
BRCA3_variant_50	4.0	2e-06	2e-05	0.94

BRCA1_variant_51	4.01	2e-06	2e-05	0.939
BRCA2_variant_52	4.02	2e-06	1.9e-05	0.938
BRCA3_variant_53	4.03	2e-06	1.9e-05	0.937
BRCA1_variant_54	4.04	2e-06	1.9e-05	0.936
BRCA2_variant_55	4.05	2e-06	1.8e-05	0.935
BRCA3_variant_56	4.06	2e-06	1.8e-05	0.934
BRCA1_variant_57	4.07	2e-06	1.8e-05	0.933
BRCA2_variant_58	4.08	2e-06	1.7e-05	0.932
BRCA3_variant_59	4.09	2e-06	1.7e-05	0.931
BRCA1_variant_60	4.1	2e-06	1.7e-05	0.93
BRCA2_variant_61	4.11	2e-06	1.6e-05	0.929
BRCA3_variant_62	4.12	2e-06	1.6e-05	0.928
BRCA1_variant_63	4.13	2e-06	1.6e-05	0.927
BRCA2_variant_64	4.14	2e-06	1.6e-05	0.926
BRCA3_variant_65	4.15	2e-06	1.5e-05	0.925
BRCA1_variant_66	4.16	2e-06	1.5e-05	0.924
BRCA2_variant_67	4.17	1e-06	1.5e-05	0.923
BRCA3_variant_68	4.18	1e-06	1.5e-05	0.922
BRCA1_variant_69	4.19	1e-06	1.4e-05	0.921
BRCA2_variant_70	4.2	1e-06	1.4e-05	0.92
BRCA3_variant_71	4.21	1e-06	1.4e-05	0.919
BRCA1_variant_72	4.22	1e-06	1.4e-05	0.918
BRCA2_variant_73	4.23	1e-06	1.4e-05	0.917
BRCA3_variant_74	4.24	1e-06	1.4e-05	0.916
BRCA1_variant_75	4.25	1e-06	1.3e-05	0.915
BRCA2_variant_76	4.26	1e-06	1.3e-05	0.914
BRCA3_variant_77	4.27	1e-06	1.3e-05	0.913
BRCA1_variant_78	4.28	1e-06	1.3e-05	0.912
BRCA2_variant_79	4.29	1e-06	1.3e-05	0.911
BRCA3_variant_80	4.3	1e-06	1.3e-05	0.91

BRCA1_variant_81	4.31	1e-06	1.2e-05	0.909
BRCA2_variant_82	4.32	1e-06	1.2e-05	0.908
BRCA3_variant_83	4.33	1e-06	1.2e-05	0.907
BRCA1_variant_84	4.34	1e-06	1.2e-05	0.906
BRCA2_variant_85	4.35	1e-06	1.2e-05	0.905
BRCA3_variant_86	4.36	1e-06	1.2e-05	0.904
BRCA1_variant_87	4.37	1e-06	1.1e-05	0.903
BRCA2_variant_88	4.38	1e-06	1.1e-05	0.902
BRCA3_variant_89	4.39	1e-06	1.1e-05	0.901
BRCA1_variant_90	4.4	1e-06	1.1e-05	0.9
BRCA2_variant_91	4.41	1e-06	1.1e-05	0.899
BRCA3_variant_92	4.42	1e-06	1.1e-05	0.898
BRCA1_variant_93	4.43	1e-06	1.1e-05	0.897
BRCA2_variant_94	4.44	1e-06	1.1e-05	0.896
BRCA3_variant_95	4.45	1e-06	1.1e-05	0.895
BRCA1_variant_96	4.46	1e-06	1e-05	0.894
BRCA2_variant_97	4.47	1e-06	1e-05	0.893
BRCA3_variant_98	4.48	1e-06	1e-05	0.892
BRCA1_variant_99	4.49	1e-06	1e-05	0.891
BRCA2_variant_100	4.5	1e-06	1e-05	0.89
BRCA3_variant_101	4.51	1e-06	1e-05	0.889
BRCA1_variant_102	4.52	1e-06	1e-05	0.888
BRCA2_variant_103	4.53	1e-06	1e-05	0.887
BRCA3_variant_104	4.54	1e-06	1e-05	0.886
BRCA1_variant_105	4.55	1e-06	1e-05	0.885
BRCA2_variant_106	4.56	1e-06	9e-06	0.884
BRCA3_variant_107	4.57	1e-06	9e-06	0.883
BRCA1_variant_108	4.58	1e-06	9e-06	0.882
BRCA2_variant_109	4.59	1e-06	9e-06	0.881
BRCA3_variant_110	4.6	1e-06	9e-06	0.88

BRCA1_variant_111	4.61	1e-06	9e-06	0.879
BRCA2_variant_112	4.62	1e-06	9e-06	0.878
BRCA3_variant_113	4.63	1e-06	9e-06	0.877
BRCA1_variant_114	4.64	1e-06	9e-06	0.876
BRCA2_variant_115	4.65	1e-06	9e-06	0.875
BRCA3_variant_116	4.66	1e-06	9e-06	0.874
BRCA1_variant_117	4.67	1e-06	9e-06	0.873
BRCA2_variant_118	4.68	1e-06	8e-06	0.872
BRCA3_variant_119	4.69	1e-06	8e-06	0.871
BRCA1_variant_120	4.7	1e-06	8e-06	0.87
BRCA2_variant_121	4.71	1e-06	8e-06	0.869
BRCA3_variant_122	4.72	1e-06	8e-06	0.868
BRCA1_variant_123	4.73	1e-06	8e-06	0.867
BRCA2_variant_124	4.74	1e-06	8e-06	0.866
BRCA3_variant_125	4.75	1e-06	8e-06	0.865
BRCA1_variant_126	4.76	1e-06	8e-06	0.864
BRCA2_variant_127	4.77	1e-06	8e-06	0.863
BRCA3_variant_128	4.78	1e-06	8e-06	0.862
BRCA1_variant_129	4.79	1e-06	8e-06	0.861
BRCA2_variant_130	4.8	1e-06	8e-06	0.86
BRCA3_variant_131	4.81	1e-06	8e-06	0.859
BRCA1_variant_132	4.82	1e-06	8e-06	0.858
BRCA2_variant_133	4.83	1e-06	8e-06	0.857
BRCA3_variant_134	4.84	1e-06	7e-06	0.856
BRCA1_variant_135	4.85	1e-06	7e-06	0.855
BRCA2_variant_136	4.86	1e-06	7e-06	0.854
BRCA3_variant_137	4.87	1e-06	7e-06	0.853
BRCA1_variant_138	4.88	1e-06	7e-06	0.852
BRCA2_variant_139	4.89	1e-06	7e-06	0.851
BRCA3_variant_140	4.9	1e-06	7e-06	0.85

BRCA1_variant_141 4.91	1e-06	7e-06	0.849
BRCA2_variant_142 4.92	1e-06	7e-06	0.848
BRCA3_variant_143 4.93	1e-06	7e-06	0.847
BRCA1_variant_144 4.94	1e-06	7e-06	0.846
BRCA2_variant_145 4.95	1e-06	7e-06	0.845
BRCA3_variant_146 4.96	1e-06	7e-06	0.844
BRCA1_variant_147 4.97	1e-06	7e-06	0.843
BRCA2_variant_148 4.98	1e-06	7e-06	0.842
BRCA3_variant_149 4.99	1e-06	7e-06	0.841
BRCA1_variant_150 5.0	1e-06	7e-06	0.84
BRCA2_variant_151 5.01	1e-06	7e-06	0.839
BRCA3_variant_152 5.02	1e-06	7e-06	0.838
BRCA1_variant_153 5.03	1e-06	7e-06	0.837
BRCA2_variant_154 5.04	1e-06	6e-06	0.836
BRCA3_variant_155 5.05	1e-06	6e-06	0.835
BRCA1_variant_156 5.06	1e-06	6e-06	0.834
BRCA2_variant_157 5.07	1e-06	6e-06	0.833
BRCA3_variant_158 5.08	1e-06	6e-06	0.832
BRCA1_variant_159 5.09	1e-06	6e-06	0.831
BRCA2_variant_160 5.1	1e-06	6e-06	0.83
BRCA3_variant_161 5.11	1e-06	6e-06	0.829
BRCA1_variant_162 5.12	1e-06	6e-06	0.828
BRCA2_variant_163 5.13	1e-06	6e-06	0.827
BRCA3_variant_164 5.14	1e-06	6e-06	0.826
BRCA1_variant_165 5.15	1e-06	6e-06	0.825
BRCA2_variant_166 5.16	1e-06	6e-06	0.824
BRCA3_variant_167 5.17	1e-06	6e-06	0.823
BRCA1_variant_168 5.18	1e-06	6e-06	0.822
BRCA2_variant_169 5.19	1e-06	6e-06	0.821
BRCA3_variant_170 5.2	1e-06	6e-06	0.82

BRCA1_variant_171 5.21	1e-06	6e-06	0.819
BRCA2_variant_172 5.22	1e-06	6e-06	0.818
BRCA3_variant_173 5.23	1e-06	6e-06	0.817
BRCA1_variant_174 5.24	1e-06	6e-06	0.816
BRCA2_variant_175 5.25	1e-06	6e-06	0.815
BRCA3_variant_176 5.26	1e-06	6e-06	0.814
BRCA1_variant_177 5.27	1e-06	6e-06	0.813
BRCA2_variant_178 5.28	1e-06	6e-06	0.812
BRCA3_variant_179 5.29	1e-06	6e-06	0.811
BRCA1_variant_180 5.3	1e-06	6e-06	0.81
BRCA2_variant_181 5.31	1e-06	6e-06	0.809
BRCA3_variant_182 5.32	1e-06	5e-06	0.808
BRCA1_variant_183 5.33	1e-06	5e-06	0.807
BRCA2_variant_184 5.34	1e-06	5e-06	0.806
BRCA3_variant_185 5.35	1e-06	5e-06	0.805
BRCA1_variant_186 5.36	1e-06	5e-06	0.804
BRCA2_variant_187 5.37	1e-06	5e-06	0.803
BRCA3_variant_188 5.38	1e-06	5e-06	0.802
BRCA1_variant_189 5.39	1e-06	5e-06	0.801
BRCA2_variant_190 5.4	1e-06	5e-06	0.8
BRCA3_variant_191 5.41	1e-06	5e-06	0.799
BRCA1_variant_192 5.42	1e-06	5e-06	0.798
BRCA2_variant_193 5.43	1e-06	5e-06	0.797
BRCA3_variant_194 5.44	1e-06	5e-06	0.796
BRCA1_variant_195 5.45	1e-06	5e-06	0.795
BRCA2_variant_196 5.46	1e-06	5e-06	0.794
BRCA3_variant_197 5.47	1e-06	5e-06	0.793
BRCA1_variant_198 5.48	1e-06	5e-06	0.792
BRCA2_variant_199 5.49	1e-06	5e-06	0.791
BRCA3_variant_200 5.5	0.0	5e-06	0.79

BRCA1_variant_201 5.51	0.0	5e-06	0.789
BRCA2_variant_202 5.52	0.0	5e-06	0.788
BRCA3_variant_203 5.53	0.0	5e-06	0.787
BRCA1_variant_204 5.54	0.0	5e-06	0.786
BRCA2_variant_205 5.55	0.0	5e-06	0.785
BRCA3_variant_206 5.56	0.0	5e-06	0.784
BRCA1_variant_207 5.57	0.0	5e-06	0.783
BRCA2_variant_208 5.58	0.0	5e-06	0.782
BRCA3_variant_209 5.59	0.0	5e-06	0.781
BRCA1_variant_210 5.6	0.0	5e-06	0.78
BRCA2_variant_211 5.61	0.0	5e-06	0.779
BRCA3_variant_212 5.62	0.0	5e-06	0.778
BRCA1_variant_213 5.63	0.0	5e-06	0.777
BRCA2_variant_214 5.64	0.0	5e-06	0.776
BRCA3_variant_215 5.65	0.0	5e-06	0.775
BRCA1_variant_216 5.66	0.0	5e-06	0.774
BRCA2_variant_217 5.67	0.0	5e-06	0.773
BRCA3_variant_218 5.68	0.0	5e-06	0.772
BRCA1_variant_219 5.69	0.0	5e-06	0.771
BRCA2_variant_220 5.7	0.0	5e-06	0.77
BRCA3_variant_221 5.71	0.0	5e-06	0.769
BRCA1_variant_222 5.72	0.0	5e-06	0.768
BRCA2_variant_223 5.73	0.0	4e-06	0.767
BRCA3_variant_224 5.74	0.0	4e-06	0.766
BRCA1_variant_225 5.75	0.0	4e-06	0.765
BRCA2_variant_226 5.76	0.0	4e-06	0.764
BRCA3_variant_227 5.77	0.0	4e-06	0.763
BRCA1_variant_228 5.78	0.0	4e-06	0.762
BRCA2_variant_229 5.79	0.0	4e-06	0.761
BRCA3_variant_230 5.8	0.0	4e-06	0.76

BRCA1_variant_231 5.81	0.0	4e-06	0.759
BRCA2_variant_232 5.82	0.0	4e-06	0.758
BRCA3_variant_233 5.83	0.0	4e-06	0.757
BRCA1_variant_234 5.84	0.0	4e-06	0.756
BRCA2_variant_235 5.85	0.0	4e-06	0.755
BRCA3_variant_236 5.86	0.0	4e-06	0.754
BRCA1_variant_237 5.87	0.0	4e-06	0.753
BRCA2_variant_238 5.88	0.0	4e-06	0.752
BRCA3_variant_239 5.89	0.0	4e-06	0.751
BRCA1_variant_240 5.9	0.0	4e-06	0.75
BRCA2_variant_241 5.91	0.0	4e-06	0.749
BRCA3_variant_242 5.92	0.0	4e-06	0.748
BRCA1_variant_243 5.93	0.0	4e-06	0.747
BRCA2_variant_244 5.94	0.0	4e-06	0.746
BRCA3_variant_245 5.95	0.0	4e-06	0.745
BRCA1_variant_246 5.96	0.0	4e-06	0.744
BRCA2_variant_247 5.97	0.0	4e-06	0.743
BRCA3_variant_248 5.98	0.0	4e-06	0.742
BRCA1_variant_249 5.99	0.0	4e-06	0.741
BRCA2_variant_250 6.0	0.0	4e-06	0.74
BRCA3_variant_251 6.01	0.0	4e-06	0.739
BRCA1_variant_252 6.02	0.0	4e-06	0.738
BRCA2_variant_253 6.03	0.0	4e-06	0.737
BRCA3_variant_254 6.04	0.0	4e-06	0.736
BRCA1_variant_255 6.05	0.0	4e-06	0.735
BRCA2_variant_256 6.06	0.0	4e-06	0.734
BRCA3_variant_257 6.07	0.0	4e-06	0.733
BRCA1_variant_258 6.08	0.0	4e-06	0.732
BRCA2_variant_259 6.09	0.0	4e-06	0.731
BRCA3_variant_260 6.1	0.0	4e-06	0.73

BRCA1_variant_261 6.11	0.0	4e-06	0.729
BRCA2_variant_262 6.12	0.0	4e-06	0.728
BRCA3_variant_263 6.13	0.0	4e-06	0.727
BRCA1_variant_264 6.14	0.0	4e-06	0.726
BRCA2_variant_265 6.15	0.0	4e-06	0.725
BRCA3_variant_266 6.16	0.0	4e-06	0.724
BRCA1_variant_267 6.17	0.0	4e-06	0.723
BRCA2_variant_268 6.18	0.0	4e-06	0.722
BRCA3_variant_269 6.19	0.0	4e-06	0.721
BRCA1_variant_270 6.2	0.0	4e-06	0.72
BRCA2_variant_271 6.21	0.0	4e-06	0.719
BRCA3_variant_272 6.22	0.0	4e-06	0.718
BRCA1_variant_273 6.23	0.0	4e-06	0.717
BRCA2_variant_274 6.24	0.0	4e-06	0.716
BRCA3_variant_275 6.25	0.0	4e-06	0.715
BRCA1_variant_276 6.26	0.0	4e-06	0.714
BRCA2_variant_277 6.27	0.0	4e-06	0.713
BRCA3_variant_278 6.28	0.0	4e-06	0.712
BRCA1_variant_279 6.29	0.0	4e-06	0.711
BRCA2_variant_280 6.3	0.0	4e-06	0.71
BRCA3_variant_281 6.31	0.0	4e-06	0.709
BRCA1_variant_282 6.32	0.0	4e-06	0.708
BRCA2_variant_283 6.33	0.0	4e-06	0.707
BRCA3_variant_284 6.34	0.0	4e-06	0.706
BRCA1_variant_285 6.35	0.0	4e-06	0.705
BRCA2_variant_286 6.36	0.0	3e-06	0.704
BRCA3_variant_287 6.37	0.0	3e-06	0.703
BRCA1_variant_288 6.38	0.0	3e-06	0.702
BRCA2_variant_289 6.39	0.0	3e-06	0.701
BRCA3_variant_290 6.4	0.0	3e-06	0.7

BRCA1_variant_291 6.41	0.0	3e-06	0.699
BRCA2_variant_292 6.42	0.0	3e-06	0.698
BRCA3_variant_293 6.43	0.0	3e-06	0.697
BRCA1_variant_294 6.44	0.0	3e-06	0.696
BRCA2_variant_295 6.45	0.0	3e-06	0.695
BRCA3_variant_296 6.46	0.0	3e-06	0.694
BRCA1_variant_297 6.47	0.0	3e-06	0.693
BRCA2_variant_298 6.48	0.0	3e-06	0.692
BRCA3_variant_299 6.49	0.0	3e-06	0.691
BRCA1_variant_300 6.5	0.0	3e-06	0.69
BRCA2_variant_301 6.51	0.0	3e-06	0.689
BRCA3_variant_302 6.52	0.0	3e-06	0.688
BRCA1_variant_303 6.53	0.0	3e-06	0.687
BRCA2_variant_304 6.54	0.0	3e-06	0.686
BRCA3_variant_305 6.55	0.0	3e-06	0.685
BRCA1_variant_306 6.56	0.0	3e-06	0.684
BRCA2_variant_307 6.57	0.0	3e-06	0.683
BRCA3_variant_308 6.58	0.0	3e-06	0.682
BRCA1_variant_309 6.59	0.0	3e-06	0.681
BRCA2_variant_310 6.6	0.0	3e-06	0.68
BRCA3_variant_311 6.61	0.0	3e-06	0.679
BRCA1_variant_312 6.62	0.0	3e-06	0.678
BRCA2_variant_313 6.63	0.0	3e-06	0.677
BRCA3_variant_314 6.64	0.0	3e-06	0.676
BRCA1_variant_315 6.65	0.0	3e-06	0.675
BRCA2_variant_316 6.66	0.0	3e-06	0.674
BRCA3_variant_317 6.67	0.0	3e-06	0.673
BRCA1_variant_318 6.68	0.0	3e-06	0.672
BRCA2_variant_319 6.69	0.0	3e-06	0.671
BRCA3_variant_320 6.7	0.0	3e-06	0.67

BRCA1_variant_321 6.71	0.0	3e-06	0.669
BRCA2_variant_322 6.72	0.0	3e-06	0.668
BRCA3_variant_323 6.73	0.0	3e-06	0.667
BRCA1_variant_324 6.74	0.0	3e-06	0.666
BRCA2_variant_325 6.75	0.0	3e-06	0.665
BRCA3_variant_326 6.76	0.0	3e-06	0.664
BRCA1_variant_327 6.77	0.0	3e-06	0.663
BRCA2_variant_328 6.78	0.0	3e-06	0.662
BRCA3_variant_329 6.79	0.0	3e-06	0.661
BRCA1_variant_330 6.8	0.0	3e-06	0.66
BRCA2_variant_331 6.81	0.0	3e-06	0.659
BRCA3_variant_332 6.82	0.0	3e-06	0.658
BRCA1_variant_333 6.83	0.0	3e-06	0.657
BRCA2_variant_334 6.84	0.0	3e-06	0.656
BRCA3_variant_335 6.85	0.0	3e-06	0.655
BRCA1_variant_336 6.86	0.0	3e-06	0.654
BRCA2_variant_337 6.87	0.0	3e-06	0.653
BRCA3_variant_338 6.88	0.0	3e-06	0.652
BRCA1_variant_339 6.89	0.0	3e-06	0.651
BRCA2_variant_340 6.9	0.0	3e-06	0.65
BRCA3_variant_341 6.91	0.0	3e-06	0.649
BRCA1_variant_342 6.92	0.0	3e-06	0.648
BRCA2_variant_343 6.93	0.0	3e-06	0.647
BRCA3_variant_344 6.94	0.0	3e-06	0.646
BRCA1_variant_345 6.95	0.0	3e-06	0.645
BRCA2_variant_346 6.96	0.0	3e-06	0.644
BRCA3_variant_347 6.97	0.0	3e-06	0.643
BRCA1_variant_348 6.98	0.0	3e-06	0.642
BRCA2_variant_349 6.99	0.0	3e-06	0.641
BRCA3_variant_350 7.0	0.0	3e-06	0.64

BRCA1_variant_351 7.01	0.0	3e-06	0.639
BRCA2_variant_352 7.02	0.0	3e-06	0.638
BRCA3_variant_353 7.03	0.0	3e-06	0.637
BRCA1_variant_354 7.04	0.0	3e-06	0.636
BRCA2_variant_355 7.05	0.0	3e-06	0.635
BRCA3_variant_356 7.06	0.0	3e-06	0.634
BRCA1_variant_357 7.07	0.0	3e-06	0.633
BRCA2_variant_358 7.08	0.0	3e-06	0.632
BRCA3_variant_359 7.09	0.0	3e-06	0.631
BRCA1_variant_360 7.1	0.0	3e-06	0.63
BRCA2_variant_361 7.11	0.0	3e-06	0.629
BRCA3_variant_362 7.12	0.0	3e-06	0.628
BRCA1_variant_363 7.13	0.0	3e-06	0.627
BRCA2_variant_364 7.14	0.0	3e-06	0.626
BRCA3_variant_365 7.15	0.0	3e-06	0.625
BRCA1_variant_366 7.16	0.0	3e-06	0.624
BRCA2_variant_367 7.17	0.0	3e-06	0.623
BRCA3_variant_368 7.18	0.0	3e-06	0.622
BRCA1_variant_369 7.19	0.0	3e-06	0.621
BRCA2_variant_370 7.2	0.0	3e-06	0.62
BRCA3_variant_371 7.21	0.0	3e-06	0.619
BRCA1_variant_372 7.22	0.0	3e-06	0.618
BRCA2_variant_373 7.23	0.0	3e-06	0.617
BRCA3_variant_374 7.24	0.0	3e-06	0.616
BRCA1_variant_375 7.25	0.0	3e-06	0.615
BRCA2_variant_376 7.26	0.0	3e-06	0.614
BRCA3_variant_377 7.27	0.0	3e-06	0.613
BRCA1_variant_378 7.28	0.0	3e-06	0.612
BRCA2_variant_379 7.29	0.0	3e-06	0.611
BRCA3_variant_380 7.3	0.0	3e-06	0.61

BRCA1_variant_381 7.31	0.0	3e-06	0.609
BRCA2_variant_382 7.32	0.0	3e-06	0.608
BRCA3_variant_383 7.33	0.0	3e-06	0.607
BRCA1_variant_384 7.34	0.0	3e-06	0.606
BRCA2_variant_385 7.35	0.0	3e-06	0.605
BRCA3_variant_386 7.36	0.0	3e-06	0.604
BRCA1_variant_387 7.37	0.0	3e-06	0.603
BRCA2_variant_388 7.38	0.0	3e-06	0.602
BRCA3_variant_389 7.39	0.0	3e-06	0.601
BRCA1_variant_390 7.4	0.0	3e-06	0.6
BRCA2_variant_391 7.41	0.0	3e-06	0.599
BRCA3_variant_392 7.42	0.0	3e-06	0.598
BRCA1_variant_393 7.43	0.0	3e-06	0.597
BRCA2_variant_394 7.44	0.0	3e-06	0.596
BRCA3_variant_395 7.45	0.0	3e-06	0.595
BRCA1_variant_396 7.46	0.0	3e-06	0.594
BRCA2_variant_397 7.47	0.0	3e-06	0.593
BRCA3_variant_398 7.48	0.0	3e-06	0.592
BRCA1_variant_399 7.49	0.0	3e-06	0.591
BRCA2_variant_400 7.5	0.0	3e-06	0.59
BRCA3_variant_401 7.51	0.0	2e-06	0.589
BRCA1_variant_402 7.52	0.0	2e-06	0.588
BRCA2_variant_403 7.53	0.0	2e-06	0.587
BRCA3_variant_404 7.54	0.0	2e-06	0.586
BRCA1_variant_405 7.55	0.0	2e-06	0.585
BRCA2_variant_406 7.56	0.0	2e-06	0.584
BRCA3_variant_407 7.57	0.0	2e-06	0.583
BRCA1_variant_408 7.58	0.0	2e-06	0.582
BRCA2_variant_409 7.59	0.0	2e-06	0.581
BRCA3_variant_410 7.6	0.0	2e-06	0.58

BRCA1_variant_411 7.61	0.0	2e-06	0.579
BRCA2_variant_412 7.62	0.0	2e-06	0.578
BRCA3_variant_413 7.63	0.0	2e-06	0.577
BRCA1_variant_414 7.64	0.0	2e-06	0.576
BRCA2_variant_415 7.65	0.0	2e-06	0.575
BRCA3_variant_416 7.66	0.0	2e-06	0.574
BRCA1_variant_417 7.67	0.0	2e-06	0.573
BRCA2_variant_418 7.68	0.0	2e-06	0.572
BRCA3_variant_419 7.69	0.0	2e-06	0.571
BRCA1_variant_420 7.7	0.0	2e-06	0.57
BRCA2_variant_421 7.71	0.0	2e-06	0.569
BRCA3_variant_422 7.72	0.0	2e-06	0.568
BRCA1_variant_423 7.73	0.0	2e-06	0.567
BRCA2_variant_424 7.74	0.0	2e-06	0.566
BRCA3_variant_425 7.75	0.0	2e-06	0.565
BRCA1_variant_426 7.76	0.0	2e-06	0.564
BRCA2_variant_427 7.77	0.0	2e-06	0.563
BRCA3_variant_428 7.78	0.0	2e-06	0.562
BRCA1_variant_429 7.79	0.0	2e-06	0.561
BRCA2_variant_430 7.8	0.0	2e-06	0.56
BRCA3_variant_431 7.81	0.0	2e-06	0.559
BRCA1_variant_432 7.82	0.0	2e-06	0.558
BRCA2_variant_433 7.83	0.0	2e-06	0.557
BRCA3_variant_434 7.84	0.0	2e-06	0.556
BRCA1_variant_435 7.85	0.0	2e-06	0.555
BRCA2_variant_436 7.86	0.0	2e-06	0.554
BRCA3_variant_437 7.87	0.0	2e-06	0.553
BRCA1_variant_438 7.88	0.0	2e-06	0.552
BRCA2_variant_439 7.89	0.0	2e-06	0.551
BRCA3_variant_440 7.9	0.0	2e-06	0.55

BRCA1_variant_441 7.91	0.0	2e-06	0.549
BRCA2_variant_442 7.92	0.0	2e-06	0.548
BRCA3_variant_443 7.93	0.0	2e-06	0.547
BRCA1_variant_444 7.94	0.0	2e-06	0.546
BRCA2_variant_445 7.95	0.0	2e-06	0.545
BRCA3_variant_446 7.96	0.0	2e-06	0.544
BRCA1_variant_447 7.97	0.0	2e-06	0.543
BRCA2_variant_448 7.98	0.0	2e-06	0.542
BRCA3_variant_449 7.99	0.0	2e-06	0.541
BRCA1_variant_450 8.0	0.0	2e-06	0.54
BRCA2_variant_451 8.01	0.0	2e-06	0.539
BRCA3_variant_452 8.02	0.0	2e-06	0.538
BRCA1_variant_453 8.03	0.0	2e-06	0.537
BRCA2_variant_454 8.04	0.0	2e-06	0.536
BRCA3_variant_455 8.05	0.0	2e-06	0.535
BRCA1_variant_456 8.06	0.0	2e-06	0.534
BRCA2_variant_457 8.07	0.0	2e-06	0.533
BRCA3_variant_458 8.08	0.0	2e-06	0.532
BRCA1_variant_459 8.09	0.0	2e-06	0.531
BRCA2_variant_460 8.1	0.0	2e-06	0.53
BRCA3_variant_461 8.11	0.0	2e-06	0.529
BRCA1_variant_462 8.12	0.0	2e-06	0.528
BRCA2_variant_463 8.13	0.0	2e-06	0.527
BRCA3_variant_464 8.14	0.0	2e-06	0.526
BRCA1_variant_465 8.15	0.0	2e-06	0.525
BRCA2_variant_466 8.16	0.0	2e-06	0.524
BRCA3_variant_467 8.17	0.0	2e-06	0.523
BRCA1_variant_468 8.18	0.0	2e-06	0.522
BRCA2_variant_469 8.19	0.0	2e-06	0.521
BRCA3_variant_470 8.2	0.0	2e-06	0.52

BRCA1_variant_471 8.21	0.0	2e-06	0.519
BRCA2_variant_472 8.22	0.0	2e-06	0.518
BRCA3_variant_473 8.23	0.0	2e-06	0.517
BRCA1_variant_474 8.24	0.0	2e-06	0.516
BRCA2_variant_475 8.25	0.0	2e-06	0.515
BRCA3_variant_476 8.26	0.0	2e-06	0.514
BRCA1_variant_477 8.27	0.0	2e-06	0.513
BRCA2_variant_478 8.28	0.0	2e-06	0.512
BRCA3_variant_479 8.29	0.0	2e-06	0.511
BRCA1_variant_480 8.3	0.0	2e-06	0.51
BRCA2_variant_481 8.31	0.0	2e-06	0.509
BRCA3_variant_482 8.32	0.0	2e-06	0.508
BRCA1_variant_483 8.33	0.0	2e-06	0.507
BRCA2_variant_484 8.34	0.0	2e-06	0.506
BRCA3_variant_485 8.35	0.0	2e-06	0.505
BRCA1_variant_486 8.36	0.0	2e-06	0.504
BRCA2_variant_487 8.37	0.0	2e-06	0.503
BRCA3_variant_488 8.38	0.0	2e-06	0.502
BRCA1_variant_489 8.39	0.0	2e-06	0.501
BRCA2_variant_490 8.4	0.0	2e-06	0.5
BRCA3_variant_491 8.41	0.0	2e-06	0.499
BRCA1_variant_492 8.42	0.0	2e-06	0.498
BRCA2_variant_493 8.43	0.0	2e-06	0.497
BRCA3_variant_494 8.44	0.0	2e-06	0.496
BRCA1_variant_495 8.45	0.0	2e-06	0.495
BRCA2_variant_496 8.46	0.0	2e-06	0.494
BRCA3_variant_497 8.47	0.0	2e-06	0.493
BRCA1_variant_498 8.48	0.0	2e-06	0.492
BRCA2_variant_499 8.49	0.0	2e-06	0.491
BRCA3_variant_500 8.5	0.0	2e-06	0.49

6.2 Biological Significance

The identification of the BRCA variant families in the top DEGs strongly correlates with existing literature on breast cancer transcriptomics, validating the mathematical accuracy of our in-browser engine compared to standard R DESeq2 pipelines.

Chapter 7: Conclusion & Future Scope

7.1 Conclusion

This dissertation successfully proves that edge-computing bioinformatics is not only possible but highly practical. BioInsight 360 achieves its primary goal: democratizing RNA-Seq analysis through an intuitive GUI while guaranteeing absolute, zero-server data privacy.

7.2 Future Scope

1. **WebAssembly Integration:** Porting the C++ source code of tools like `kallisto` into WebAssembly to allow raw FASTQ read pseudoalignment in the browser.
2. **IndexedDB for Gigabyte Scaling:** Moving from `localStorage` (5MB limit) to IndexedDB (Gigabyte limit) to allow the processing of massive single-cell RNA-Seq (scRNA-Seq) datasets.

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